

Practicum CFD 2026-2027

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Practicum 4

deadline: Thursday, October 8th, 17:00

- Working in pairs (groups of max. 2 students) is allowed.
- Each practicum is closed with an oral discussion supported by a (short) written report of the findings. Make the report by pen/paper (readable) or with L^AT_EX, Word, etc.
- Always use Python or Matlab/Octave for the programs that you develop.
- Save your programs in separate files, such that it is easy to demonstrate them.
- The use of AI tools such as ChatGPT is not allowed!
- Be sure that you have discussed the exercises and programs before the deadline!

Exercise 1 – Implement unsteady term

”Copy and Modify” the program you composed in Practica 1 and 2 to solve the unsteady convection-diffusion equation

$$\frac{d\phi}{dt} + U \frac{d\phi}{dx} = k \frac{d^2\phi}{dx^2} \quad \text{on } 0 \leq x \leq L, \quad \text{with } \phi(0) = \phi_0, \quad \phi(L) = \phi_1.$$

Take into account the following:

- Both explicit and implicit 1st order time discretisation should be possible.
- The time discretisations should be possible in combination with:
 - central and 1st order upwind spatial discretisation on equidistant grids
 - method A for variable grids.
- Very fine grids, both in spatial direction and in time(!), should be possible.

Exercise 2 – Stability study

- Analyse the stability of Method A, in combination with **explicit** time discretisation, on a spatial grid with constant stretch rate: $h + /h- = 0.95$.
 - Determine the stability requirements (pos. coeff.) analytically, and check whether these can be confirmed by means of simulations with your program.
 - Are the requirements ”worse” than those for a constant grid spacing?
- Analyse the stability of Method A, in combination with **implicit** time discretisation, on a spatial grid with constant stretch rate: $h + /h- = 0.95$.

Try to determine the stability requirements (pos. coeff.) analytically, and check whether these can be confirmed by means of simulations.
- Modify your code: implement the initial condition $\phi(t=0) = 2 + \sin(\frac{\pi}{L}x)$ and periodic conditions $\phi(0) = \phi(L) = 2$. What is the steady solution now?
 - Check the stability requirements $\eta < d < 1$ (pos. coeff.) and $\eta^2 < d < 1$ (Fourier) by doing simulations on an equidistantly spaced grid, using central and explicit discretisation. Which requirements hold in this case?

Exercise 3 – Wave simulation

Set $k = 0$ (zero diffusion), and choose explicit discretisation in combination with upwind discretisation on an equidistantly spaced grid that covers $[0 ; 1]$.

Implement the initial condition $\phi(t=0) = 2 + \sin(6\pi x)$, together with boundary conditions $\phi(x=0) = 2 + \sin(6\pi(0 - Ut))$ and $\phi(x=1) = 2 + \sin(6\pi(1 - Ut))$.

- Check whether the stability requirement $\eta \leq 1$ can be confirmed by means of simulations with your program.
- Which η -value should be taken to suppress abundant (numerical) damping? Check whether this can be confirmed by means of simulations.

Exercise 4 – Unsteady diffusion and stability

Description

Consider the heat equation (pure diffusion, zero convection, $U = 0$)

$$\frac{\partial T}{\partial t} = \frac{\partial^2 T}{\partial x^2}, \quad 0 \leq x \leq 1, \quad 0 \leq t \leq 4,$$

with boundary conditions $T(0, t) = \frac{\partial T}{\partial x}(1, t) = 0$ and initial condition $T(x, 0) = x$.

This equation is solved with a finite-difference method on a grid with N grid points ($N = 10$ or 20). The Neumann condition is discretized with a mirror point.

As time-integration method the generalized Crank-Nicolson method is used

$$\frac{T_j^{n+1} - T_j^n}{\delta t} = (1 - \omega) \frac{T_{j+1}^n - 2T_j^n + T_{j-1}^n}{h^2} + \omega \frac{T_{j+1}^{n+1} - 2T_j^{n+1} + T_{j-1}^{n+1}}{h^2}.$$

This gives for $\omega = 0$ the fully explicit method, for $\omega = 0.5$ the Crank-Nicolson method and for $\omega = 1$ the fully implicit method.

Required files: `cf4.m`.

Typing the command `cf4` in Matlab asks for the input of the required parameters for this exercise: N , ω (omega) and δt . The quantity $T(1, t)$ is monitored.

Questions to be solved on the computer

- First consider the fully explicit method ($\omega = 0$). Vary the number of grid points as $N = 10$ and $N = 20$. Solve the heat equation for various time steps, as indicated in the table below. Monitor whether (yes/no) the simulation is stable. In (d) you will be asked to explain the stability limit on the time step.

time step	0.001 stable?	0.00125 stable?	0.00126 stable?	0.0025 stable?	0.005 stable?	0.0051 stable?	0.01 stable?
N=10							
N=20							

- Next, use the Crank-Nicolson method ($\omega = 0.5$) and the generalized Crank-Nicolson method with $\omega = 0.6$. Use $N = 10$. Set the time step δt at 0.05 and 0.5; see the table below. For the larger time step the solution shows clear oscillations. Make a rough estimate for the damping factor (defined as the quotient of two successive amplitudes) of the oscillations.

time step	$\delta t = 0.05$ stable?	$\delta t = 0.5$	
		stable?	damping
$\omega = 0.5$			
$\omega = 0.6$			

- (c) Finally, again for $N = 10$, investigate the fully implicit method ($\omega = 1$). Use the same time steps as in (b). Observe whether the solution shows oscillations.

time step	$\delta t = 0.05$		$\delta t = 0.5$	
	stable?	oscillations?	stable?	oscillations?
$\omega = 1.0$				

Questions to be solved by pencil-and-paper

- (d) First consider the fully explicit method ($\omega = 0$). Carry out a Fourier analysis of this method, by using $\eta = 0$ in the formulas of the reader. Determine the maximum allowable time step for $N = 10$ and $N = 20$. Compare these stability limits with the empirical observations in (a).
- (e) Carry out a Fourier stability analysis of the generalized Crank-Nicolson method, and determine the amplification factor. Show that the generalized Crank-Nicolson method is unconditionally stable for $\omega \geq 0.5$. Investigate how the amplification factor behaves when $\delta t \rightarrow \infty$. Now explain the oscillations visible in (b). How are these oscillations influenced when ω is increased from 0.5 to 0.6? Give a possible explanation why the observed amplification factors in (b) are somewhat different from the theoretical amplification factors.
- (f) Derive theoretically for which values of the Crank-Nicolson parameter ω the solution is wiggle-free. As a special case, explain why the fully implicit method ($\omega = 1$) does not show oscillations, see (c). Hint: use the concept of positive coefficients.
- (Bonus) This is extra. This exercise is not an official part of the practicum. Try to derive the stability requirements for the unsteady convection-diffusion equation

$$\frac{d\phi}{dt} + U \frac{d\phi}{dx} = k \frac{d^2\phi}{dx^2} \quad \text{on } 0 \leq x \leq L, \quad \text{with } \phi(0) = \phi_0, \quad \phi(L) = \phi_1.$$

depending on the values of the Crank-Nicolson parameter ω . The spatial derivatives are treated with central discretisation, the time derivative with the generalized Crank-Nicolson method.